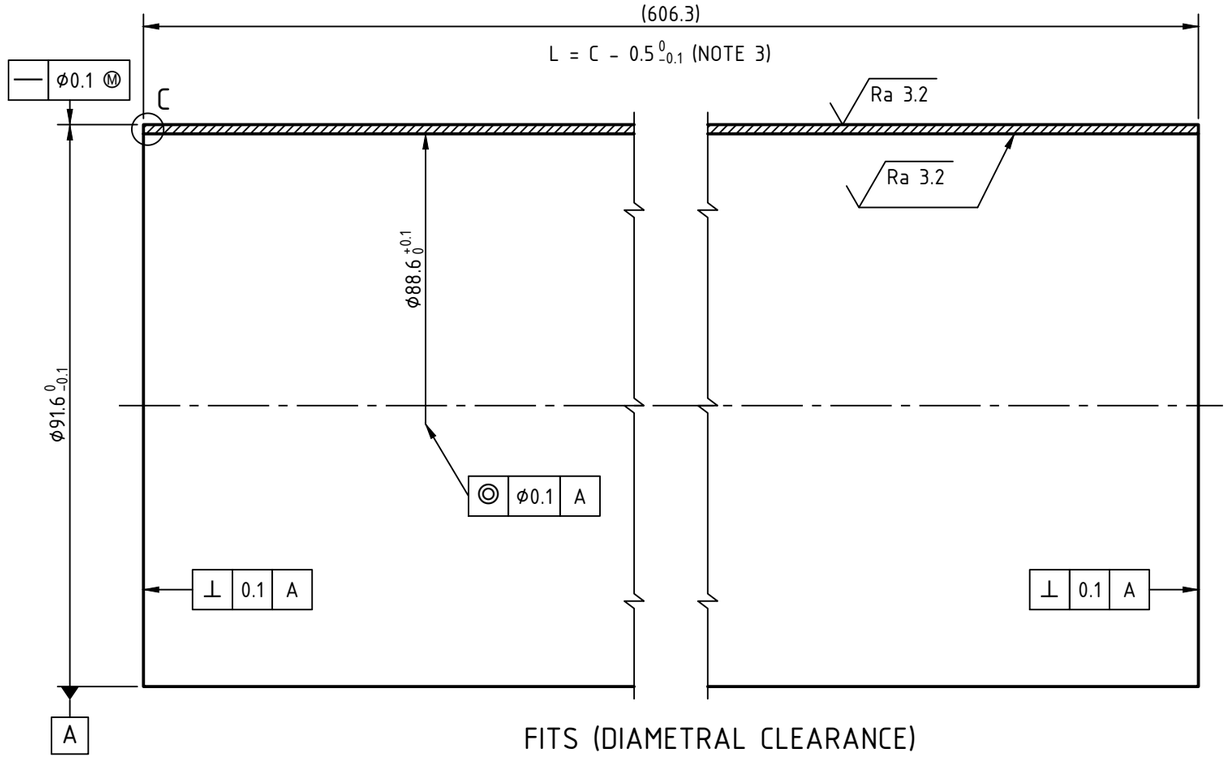


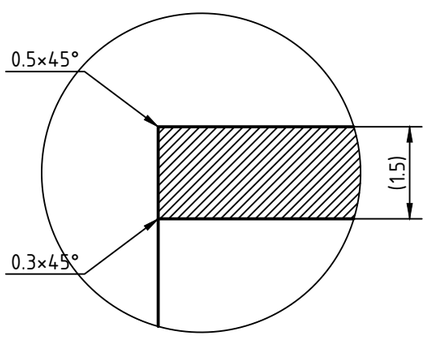
NOTES

1. Material: phenolic-impregnated kraft paper tube, convolute or spiral wound; paper-base phenolic to NEMA LI 1 grade X / XX or equal, density approx. 1.35 g/cm³. Stock: tube bought to the drawn OD and bore, or OD ground / bore honed from an oversize tube; supply length ≥ 612. Free of delamination, blisters, soft spots and cracks. Certificate of conformity required.
2. Datum A is the OD axis. Sizes are two-point sizes at 20 °C on a tube conditioned dry (sealed with desiccant ≥ 24 h): paper phenolic swells with humidity. Store sealed.
3. Length L is cut at assembly to the measured cavity C, nozzle-insulator forward face to forward-insulator aft face (SRM100-000): L = C − 0.5 0/−0.1. (606.3) is L at the nominal C = 606.8. The liner carries no axial load: before the forward closure goes in, the depth from the case forward end face to the liner forward end must be ≥ t + 0.30 at 4 points, t = closure flange bearing face to forward-insulator aft face (SRM100-000 sheet 3, note 8).
4. Edges drawn sharp. Break both OD end edges 0.5x45° (protects the honed seal zones of SRM100-101 when the liner is pushed in) and both bore end edges 0.3x45°. Cut ends clean: no loose fibres, no crushed or split ends.
5. Clean and dry inside and out before assembly. Grease the OD at assembly: Parker Super O-Lube or DC-55, thin film; no grease in the bore. Must slide by hand through case SRM100-101 over its full length. Casting tubes, spacers and take-up ring must slide freely in the bore.
6. Inspect 100 %: OD and bore at both ends and mid-length, 2 directions 90° apart; wall ≥ 1.35 at 4 positions each end; straightness ⌀0.1 Ⓜ: a full-length sleeve gauge ⌀91.70 passes over the OD by hand; length at 4 positions 90° apart after cutting. Visual inside and out. Record with the case serial.
7. Mark SRM100-110, batch and the serial of the case it is cut for on the OD, within 30 of one end: indelible ink, 3 mm characters. Do not engrave or stamp.
8. Installed from motor z 43 (bears on nozzle insulator SRM100-105) to z 649.3; the forward end floats 0.5 short of forward insulator SRM100-107.
9. Used on SRM100-000 item 8 (1 off). Single use: replaced after every firing (README, After every firing).

FITS (DIAMETRAL CLEARANCE)

MATING PART	BORE	SHAFT	CLEARANCE
Liner OD in case bore SRM100-101, full length	⌀92.000 … 92.10	⌀91.50 … 91.60	0.40 … 0.60
Casting tube SRM100-111 in liner bore	⌀88.60 … 88.70	⌀88.30 … 88.40	0.20 … 0.40
Grain spacer SRM100-109 in liner bore	⌀88.60 … 88.70	⌀88.30 … 88.40	0.20 … 0.40
Take-up ring SRM100-108 in liner bore	⌀88.60 … 88.70	⌀88.00 … 88.40	0.20 … 0.70

Both liner ends lie in the honed seal zones of SRM100-101 (45 +5/0 from each case end face: overlap ≥ 2 aft, ≥ 3.5 fwd), held to A +0.030/0 with 92.000 ≤ A ≤ 92.070, A recorded per zone: the bore is ≤ 92.10 everywhere, zones included, so the clearance 0.40 … 0.60 holds over the full length. Straightness ⌀0.1 Ⓜ: liner virtual size ⌀91.70 = smallest straight case bore, GO plug ⌀91.70 (bore min 92.000, surface lines straight within 0.3, SRM100-101).



C (10:1)

Ra 6.3 (√)

MATERIAL: Paper-base phenolic tube, NEMA LI 1 grade X / XX or equal	MASS: 347.5 g	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: Ra 6.3	EDGES: ISO 13715 −0.3 / +0.3
DOCUMENT TYPE: DETAIL DRAWING	DOCUMENT STATUS: NOT CHECKED - FOR REVIEW	SCALE: 1:1	PROJECTION	UNITS: mm
LEGAL OWNER: HELIOS, SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: LINER — Case liner / thermal barrier, ⌀91.6 / ⌀88.6 - paper phenolic			
CREATED BY: text-to-cad / Claude	APPROVED BY: -	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-110	REV: A / SHEET: 1 / 1